Uncertainty intervals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

State the model. Calculate the bounds. Explain the coverage.

Compute a known-sigma normal mean interval and interpret separate supplied differences without inventing missing data or stronger claims.

Keep one habit: Interval contract → standard-error/margin ledger → supplied-difference native plot → target/model probes → complete numeric audit → independent new inputs.

MAIN LESSON · SECTION 1

Calculate a mean interval under a stated sampling model

> Separate arithmetic, confidence-method meaning and unverified assumptions.

Learning goals

I can explain the **mechanism** and terminology of uncertainty intervals.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve sampling variability, units and known-model probabilities

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Sampling variability. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain uncertainty intervals using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of uncertainty intervals

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define the target, formula and confidence meaning before calculating

> The interval method depends on the sampling assumptions, not just the summary numbers.

Begin with a mean and its uncertainty

A sample mean is one estimate. An interval-producing method describes its uncertainty under specified assumptions. In this exercise, sixteen independent normal measurements have observed mean ten and known population standard deviation two. Compute a standard error, multiply by the supplied critical value and extend the interval around the mean.

The confidence level describes repeated use of the procedure under its model; it is not a new probability that this fixed unknown mean lies in the already observed interval.

Define confidence-method interpretation

An uncertainty interval needs a sampling model and assumptions. A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

Complete known-sigma normal-model contract

Interval item	Complete constructed-exercise contract
Target	Population mean, not the range of individual observations
Sampling assumptions	Independent measurements from a normal population; known positive population standard deviation
Main supplied summaries	Observed mean 10 units; sigma 2 units; n=16
Standard error of sample mean	SE = sigma / sqrt(n)
Supplied critical value	1.96, rounded normal critical value for approximately 95% coverage
Margin and endpoints	Margin =1.96×SE; interval = mean ± margin
Confidence interpretation	Long-run interval-procedure coverage under assumptions; fixed-parameter probability not supplied
Known sigma versus estimated spread	Do not substitute sample SD into this exact known-sigma formula without a justified method
Model contrast	This independent-normal model differs from the previous finite-population without-replacement exercise
Limits	Numeric guards cannot verify independence, normality, known sigma or unbiased sampling; interval does not repair bias or establish practical value/causality

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace standard error, margin and both endpoints

> Keep population spread sigma distinct from the standard error of the mean.

Original independent-normal known-sigma procedure

For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96: mean plus/minus 1.96*sigma/sqrt(n). Un**known sigma** requires a different justified method.

Complete arithmetic and unit ledger

Main calculation step	Substitution	Result and unit
Root sample size	sqrt(16)	4, dimensionless
Standard error	sigma / sqrt(n) =2/4	0.5 measurement units
Margin	1.96 ×0.5	0.98 measurement units
Lower endpoint	10 −0.98	9.02 measurement units
Upper endpoint	10 +0.98	10.98 measurement units

Main calculation step	Substitution	Result and unit
Reported interval	[9.02, 10.98]	For the population mean under stated assumptions
What is not calculated	Raw individual-value coverage or unknown-sigma interval	No actual sampling or coverage experiment

State which assumptions the numbers cannot verify

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Uncertainty intervals — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the standard error, margin and interval.	For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96: mean plus/minus 1.96*sigma/sqrt(n). Unknown sigma requires a different justified method.	SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict a supplied difference interval that includes zero

> Zero inclusion does not prove exact equality or erase the estimate.

Current task and response

- Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict changed supplied bounds

- Change: Use a supplied interval [-1.4,0.3] instead.

▼ Check after predicting

Prediction: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval.

Reason: The new upper endpoint crosses zero.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Read estimate, comparison direction and zero against the supplied bounds

> Keep interpretation of a supplied interval separate from calculation with known sigma.

Main computed interval and checked result

Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval.

SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate time-difference inputs and method assumption

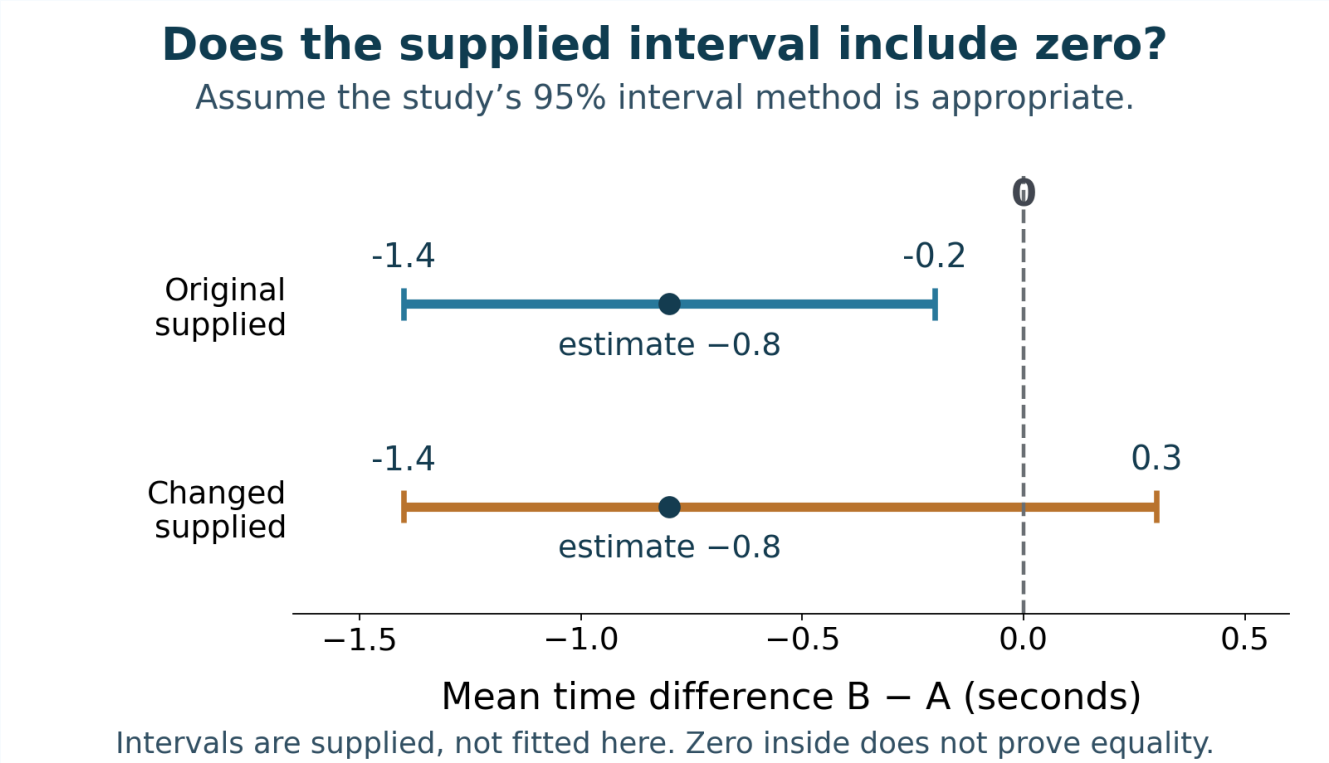
Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

The picture is a separate time-difference report. Its estimate B–A is –0.8 seconds and its supplied 95% interval is [–1.4, –0.2], assuming the study's method is appropriate. All endpoints are negative, so the interval supports a negative mean-time difference under that method, not a claim that every request is faster.

The changed supplied interval [–1.4, 0.3] includes zero and both negative/positive differences; it does not establish a definite direction or prove equality. Raw data, sample size and fitting method are absent. Do not derive its standard error, force symmetric bounds or apply the main known-sigma formula to it. Practical value and causality need additional evidence.

See original and changed bounds in the native plot



AI ACADEMY / AI-7/12 · 02 · TRACE THE CASE

Follow the deciding steps



01 Locate the estimate

-0.8 lies between the interval endpoints.



02 Interpret the sign

Negative B−A means B took less time under this measurement convention.



03 Respect coverage meaning

95% describes the method’s long-run coverage, not a 95% probability for this fixed parameter.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Locate the estimate	-0.8 lies between the interval endpoints.
2. Interpret the sign	Negative B–A means B took less time under this measurement convention.
3. Respect coverage meaning	95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Compare supplied intervals without reconstructing raw data

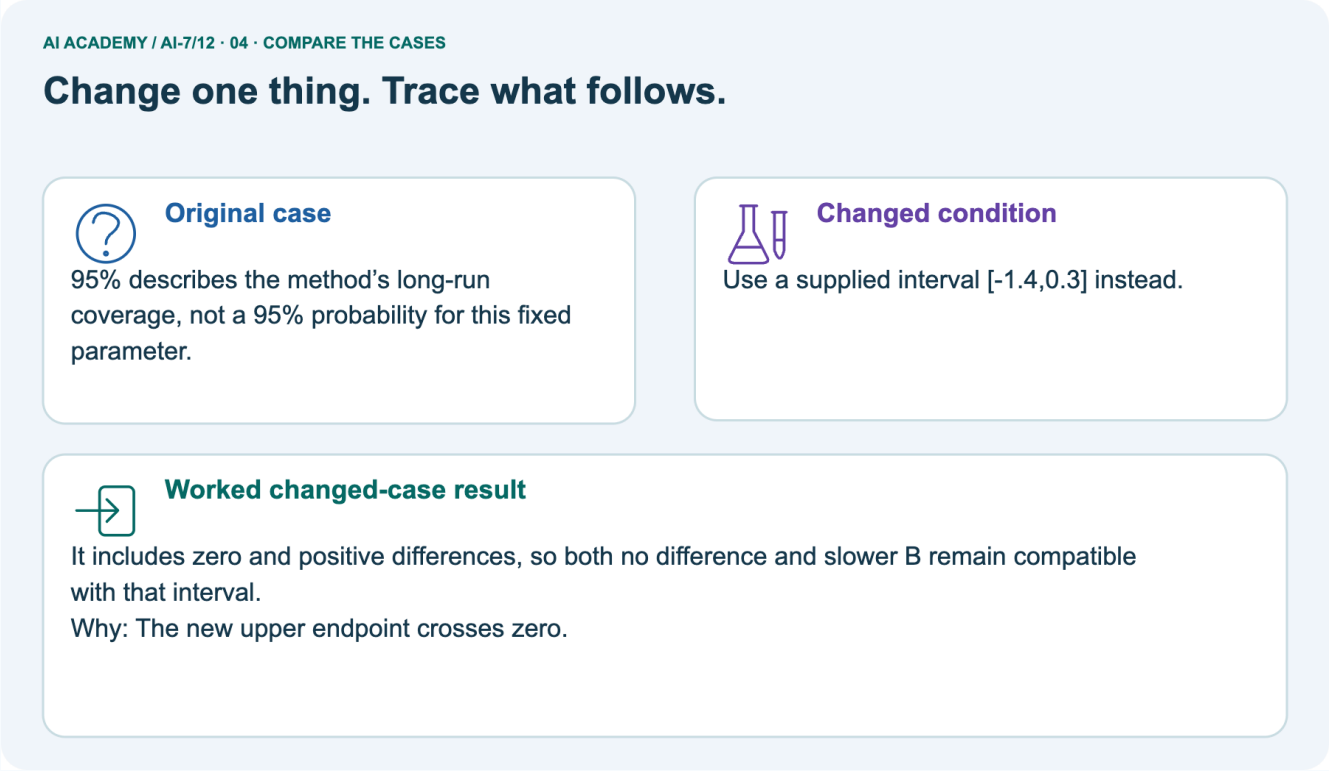
Separate supplied report	Original interval	Changed interval
Point estimate B–A	−0.8 seconds	−0.8 seconds, retained in the scenario
Bounds	[−1.4, −0.2] seconds	[−1.4, 0.3] seconds
Zero relative to bounds	Outside, to the right	Inside
Mean-difference direction	Entire supplied range negative	Both signs and zero included

Separate supplied report	Original interval	Changed interval
Valid caution	Not every request's time or a causal effect	Zero inclusion is not proof of equality
Unavailable ingredients	Raw data and interval fitting details	Still unavailable; no recalculation claimed

Predict the changed zero-including interval

- Change one thing: Use a supplied interval [-1.4,0.3] instead. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain invalid assumptions, practical value and confidence meaning

- An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

- Read the labelled picture: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

▼ Earlier wording, alternatives and picture transcripts

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from math import sqrt

def known_sigma_interval(mean, sigma, n, critical=1.96):
    """Assume independent normal sampling and a known population sigma."""
    if type(n) is not int or n < 1 or sigma < 0 or (critical <= 0):
        raise ValueError('interval settings')
    se = sigma / sqrt(n)
    margin = critical * se
    return (se, mean - margin, mean + margin)

print(known_sigma_interval(10, 2, 16))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Uncertainty intervals

The facts we will use

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Supplied interval for B minus A time

	Lower endpoint (s)	Estimate (s)	Upper endpoint (s)
Original supplied interval	-1.4	-0.8	-0.2
Changed supplied interval	-1.4	not specified	0.3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Uncertainty intervals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Locate the estimate

-0.8 lies between the interval endpoints.



2. Interpret the sign

Negative B-A means B took less time under this measurement convention.



3. Respect coverage meaning

95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

CHANGE ONE THING

Use a supplied interval [-1.4,0.3] instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit confidence meaning, target and missing assumptions

> An interval is a model-dependent analysis result rather than an all-purpose guarantee.

Core interval qualification and failure analysis

A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete target, sign and model probes

Interval probe	Expected interpretation
Known sigma 2 and n16	SE 0.5; does not mean raw observations have SD 0.5
A 95% confidence method	Approximate long-run coverage under assumptions, not 95% of individual observations
Fixed observed interval	No 95% posterior parameter probability supplied by this frequentist calculation
Unknown sigma	Choose an appropriate justified method; do not silently pretend known
Biased sampling or correlated records	Numeric interval may not have intended coverage
Zero inside a difference interval	No definite sign conclusion from these bounds; not proof of equality
Negative B–A mean interval	B lower mean time under stated convention, not every paired request faster
Supplied asymmetric interval	Keep supplied endpoints; no assumed symmetric-SE reconstruction

Interval probe	Expected interpretation
Finite positive inputs pass code guards	Arithmetic admissible; real model/design assumptions still unverified

Guided repair and explanation

- A report omits this qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise tracing units and interpreting interval signs

> State whether the interval was calculated here or only supplied.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example “Uncertainty intervals”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Uncertainty intervals. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

> **My explanation using a pictured step and the given facts**

Reserve the selected new mean/sigma/sample-size calculation

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete interval calculation and supplied-bound check

> Record arithmetic separately from untested sampling assumptions.

Current known-sigma mean-interval investigation

Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

Runnable complete numeric and supplied-bound audit

```
"""Known-sigma normal interval arithmetic and separate supplied-interval
interpretation."""
from math import sqrt,isfinite
from statistics import NormalDist

def finite_real(value):
 if type(value) not in (int,float):return False
 try:return isfinite(value)
 except OverflowError:return False

def known_sigma_interval(mean,sigma,n,critical=1.96):
 """Assume independent normal sampling with known positive population sigma.
 Numeric checks cannot verify those design/model assumptions.
 """
 if not finite_real(mean) or not finite_real(sigma) or sigma<=0:
  raise ValueError('finite mean and known positive sigma required')
 if type(n) is not int or n<1 or not finite_real(critical) or critical<=0:
  raise ValueError('positive exact n and finite positive critical required')
 try:
  se=sigma/sqrt(n);margin=critical*se;lower=mean-margin;upper=mean+margin
 except OverflowError:raise ValueError('settings exceed this floating-point calculation')
 if not all(isfinite(v) for v in [se,margin,lower,upper]):raise ValueError('nonfinite
result')
 return
{'standard_error':se,'margin':margin,'lower':lower,'upper':upper,'critical':critical,
         'theoretical_normal_coverage':2*NormalDist().cdf(critical)-1}
main=known_sigma_interval(10,2,16)
assert main['standard_error']==0.5 and abs(main['margin']-0.98)<1e-12
assert abs(main['lower']-9.02)<1e-12 and abs(main['upper']-10.98)<1e-12
assert abs(main['theoretical_normal_coverage']-0.95)<0.00001
```

```
# Separate supplied intervals; no standard error or raw-data fitting is inferred.
def locate_zero(lower,upper,estimate):
 if not all(finite_real(v) for v in [lower,upper,estimate]) or lower>upper or not
lower<=estimate<=upper:
  raise ValueError('ordered finite supplied bounds containing the estimate required')
 return {'contains_zero':lower<=0<=upper,
         'direction':'all negative' if upper<0 else 'all positive' if lower>0 else
'includes zero'}
picture=locate_zero(-1.4,-0.2,-0.8)
changed=locate_zero(-1.4,0.3,-0.8)
assert picture=={'contains_zero':False,'direction':'all negative'} and
changed['contains_zero']
print('Main known-sigma interval:',main)
print('Separate supplied B-A time interval:',picture,'changed supplied interval:',changed)
print('Negative B-A means B has lower mean time under this convention, not every request
is faster.')
print('Critical 1.96 is rounded: theoretical normal-model coverage is approximately 95
percent. No repeated-sample coverage experiment was run.')
print('Confidence concerns the procedure under assumptions, not a 95 percent probability
assigned to a fixed unknown mean after observing this interval. Zero inclusion does not
prove equality.')
print('The calculation does not verify normality, independence, known sigma or unbiased
sampling; an interval does not repair bias, prove causality or establish practical
value.')
```

Expected calculations and model limits

1. Predict standard error 0.5, margin 0.98 and interval [9.02, 10.98] before running or tracing. The code rejects nonfinite values, Boolean counts and nonpositive sigma for this nondegenerate normal exercise. Its theoretical normal coverage calculation explains why rounded 1.96 gives approximately 95%, not an observed coverage trial.

   The separate supplied difference intervals are located against zero without inferring their raw data or standard error. No normality, independence, known-sigma status or unbiased collection is verified by arithmetic. The reserved mean 20/sigma 5/n25 case remains for the first individual attempt.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier compact illustration and limited guards

```
from math import sqrt

def known_sigma_interval(mean, sigma, n, critical=1.96):
    """Assume independent normal sampling and a known population sigma."""
    if type(n) is not int or n < 1 or sigma < 0 or (critical <= 0):
        raise ValueError('interval settings')
    se = sigma / sqrt(n)
    margin = critical * se
    return (se, mean - margin, mean + margin)

print(known_sigma_interval(10, 2, 16))
```

The earlier compact reference code illustrates the default arithmetic but omits some finite-value and nondegenerate-input guards. Use the complete public investigation for this lesson’s explicit positive-sigma contract. Neither version can verify the actual sampling assumptions from summary numbers.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: calculate a new known-sigma mean interval

> Show standard error, margin, endpoints and one required assumption.

Complete selected independent task

Observed mean is 20 from 25 independent normal measurements; population standard deviation is known to be 5. Use the supplied 95% rule mean ± 1.96 × sigma / sqrt(n). Compute the interval and name one assumption.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Observed mean is 20 from 25 independent normal measurements; population standard deviation is known to be 5. Use the supplied 95% rule mean ± 1.96 × sigma / sqrt(n). Compute the interval and name one assumption.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save interval target, model, inputs, units and confidence meaning

> Keep supplied endpoints separate from independently calculated ones.

Project task and artifact

Add a labelled uncertainty intervals **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of uncertainty intervals.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local calculation or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain what the interval procedure covers and what it does not establish

> Do not turn zero inclusion, model assumptions or a confidence level into stronger claims.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of uncertainty intervals

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Standard error	The model-based variation of an estimate
Confidence interval	An interval procedure with declared repeated-sampling coverage
Known sigma	A supplied population standard deviation assumption

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Uncertainty intervals — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the standard error, margin and interval.	For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96: mean plus/minus 1.96*sigma/sqrt(n). Unknown sigma requires a different justified method.	SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

By the end, I can	How I will show it
Explain the mechanism and terminology of uncertainty intervals	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.
- Which labelled step matches this detail? 95% describes the method’s long-run coverage, not a 95% probability for this fixed parameter.
- 1. Interpret the sign
- 2. Locate the estimate
- 3. Respect coverage meaning
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Respect coverage meaning 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Changed-case question and optional hint

- Changed-case practice: Use a supplied interval [-1.4,0.3] instead. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. It includes zero and positive differences, so both no difference and slower B remain compatible with that interval.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Repair and retain the record

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Experimental units and independence (ai-7/9). Goal: Explain the **mechanism** and terminology of experimental units and independence. Short practice: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

</details>

<details>
<summary>Reference · Six-picture concept journey</summary>

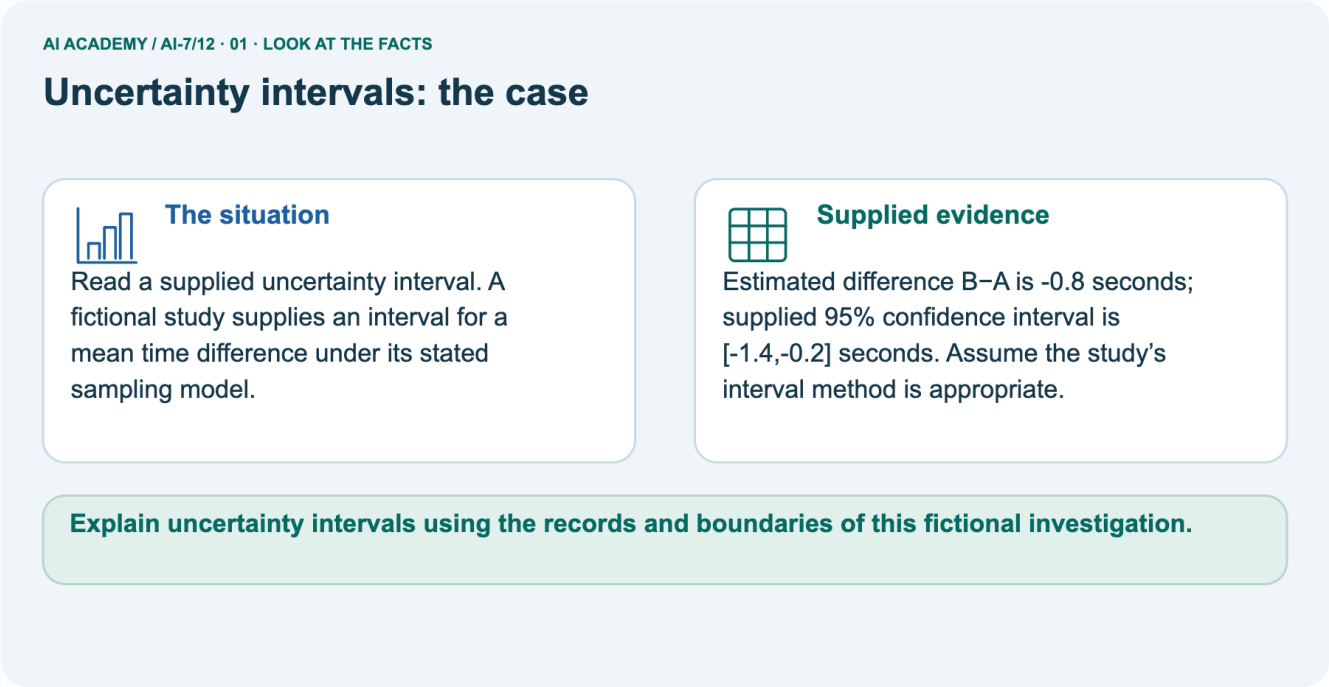
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REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Uncertainty intervals: the case



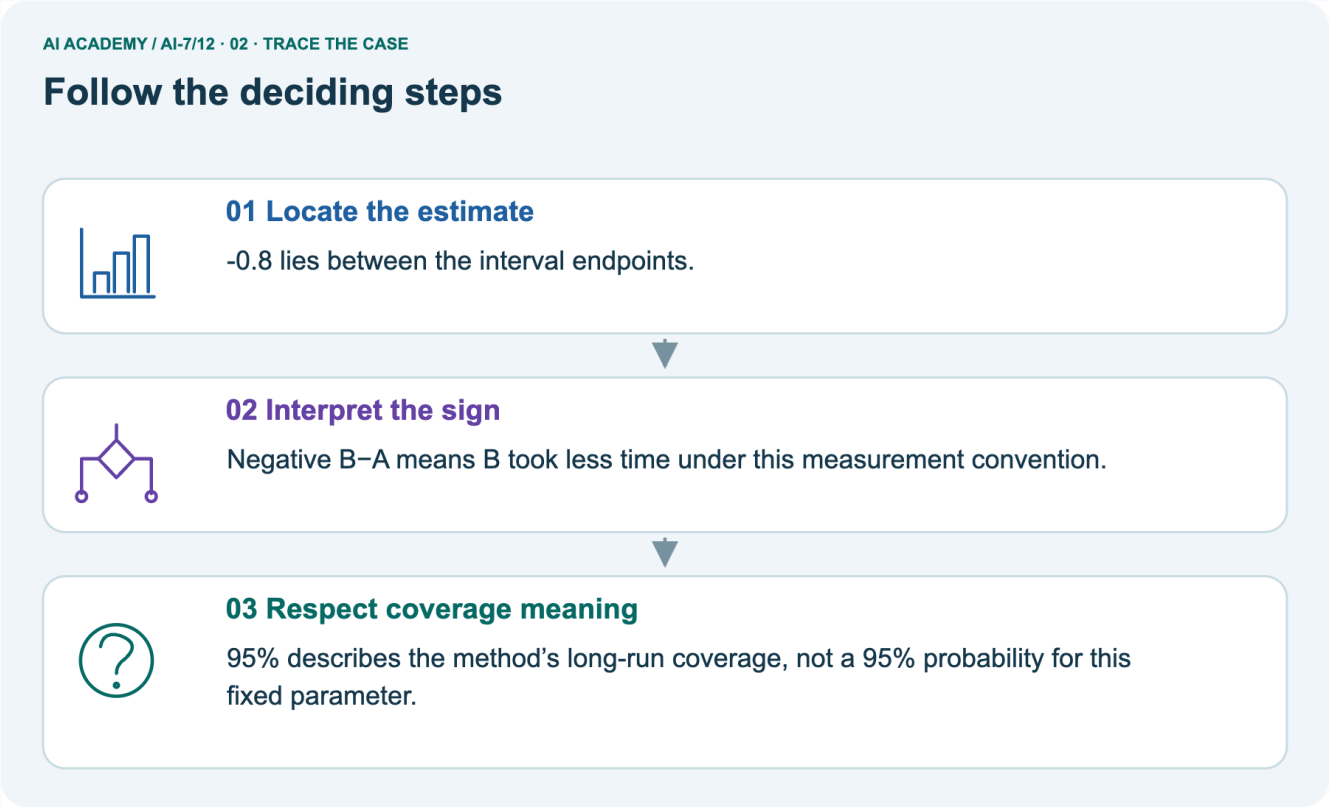
Follow the labelled picture with the matching explanation. Open the vector diagram

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Supplied facts: Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

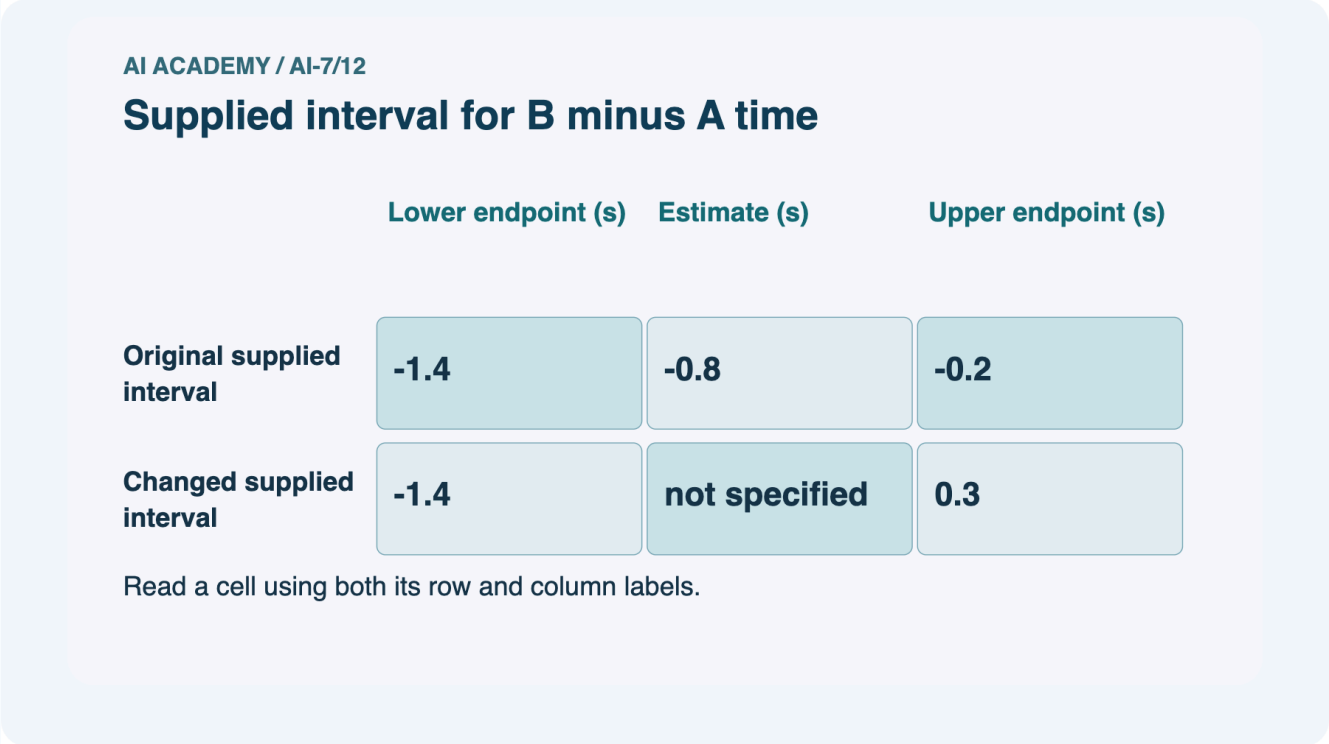


Follow the labelled picture with the matching explanation. Open the vector diagram

Locate the estimate: -0.8 lies between the interval endpoints. → Interpret the sign: Negative B–A means B took less time under this measurement convention. → Respect coverage meaning: 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Point to each numbered stage and explain which recorded **input** it uses.

03. Supplied interval for B minus A time



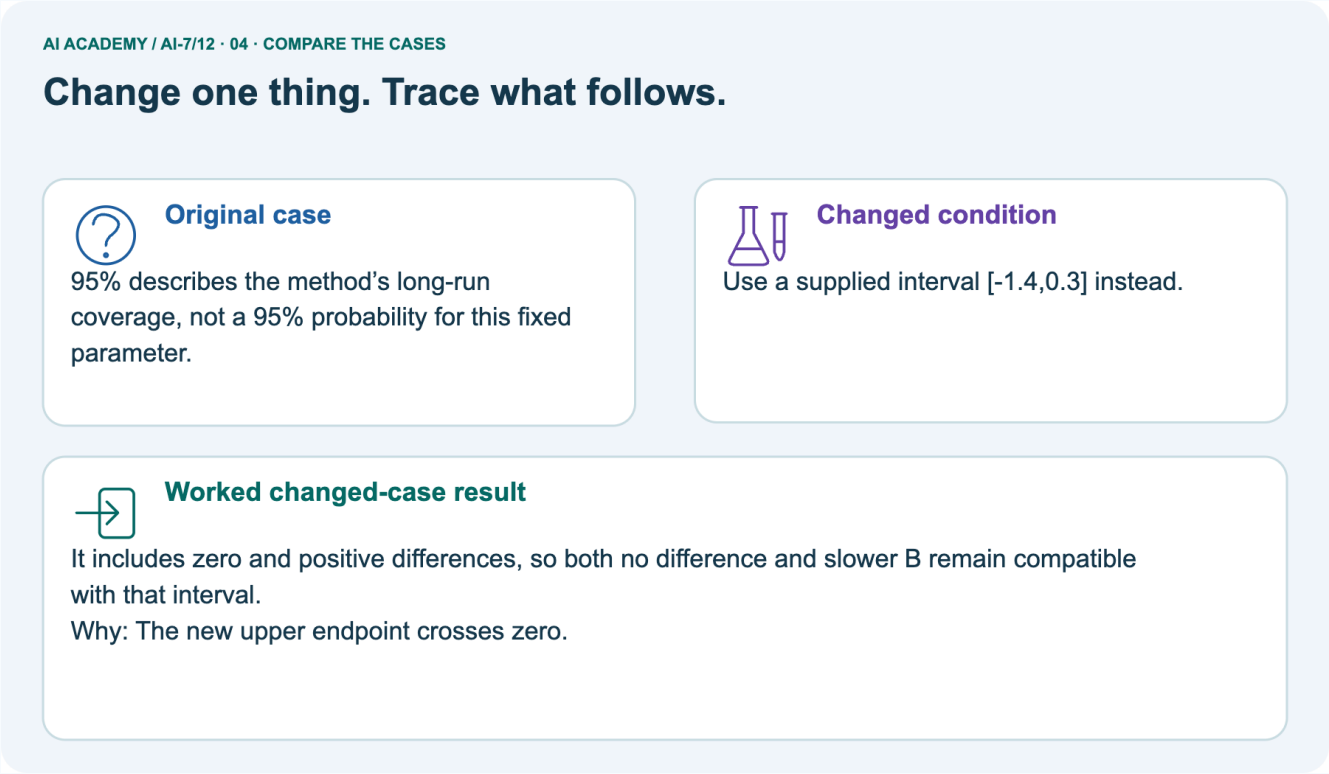
Follow the labelled picture with the matching explanation. Open the vector diagram

Supplied interval for B minus A time. Original supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = -0.8, Upper endpoint (s) = -0.2.

Changed supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = not specified, Upper endpoint (s) = 0.3.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



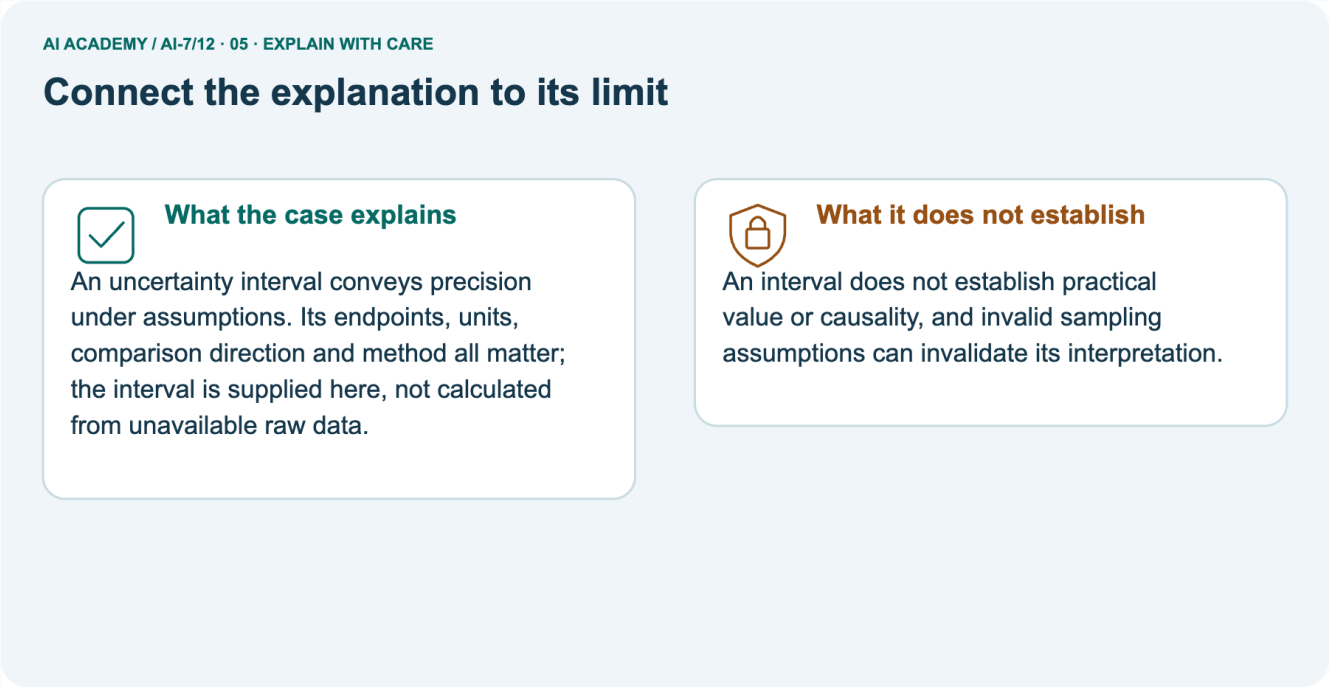
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 95% describes the method’s long-run coverage, not a 95% probability for this fixed parameter. Changed condition: Use a supplied interval [-1.4,0.3] instead.

Expected result: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. Reason: The new upper endpoint crosses zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



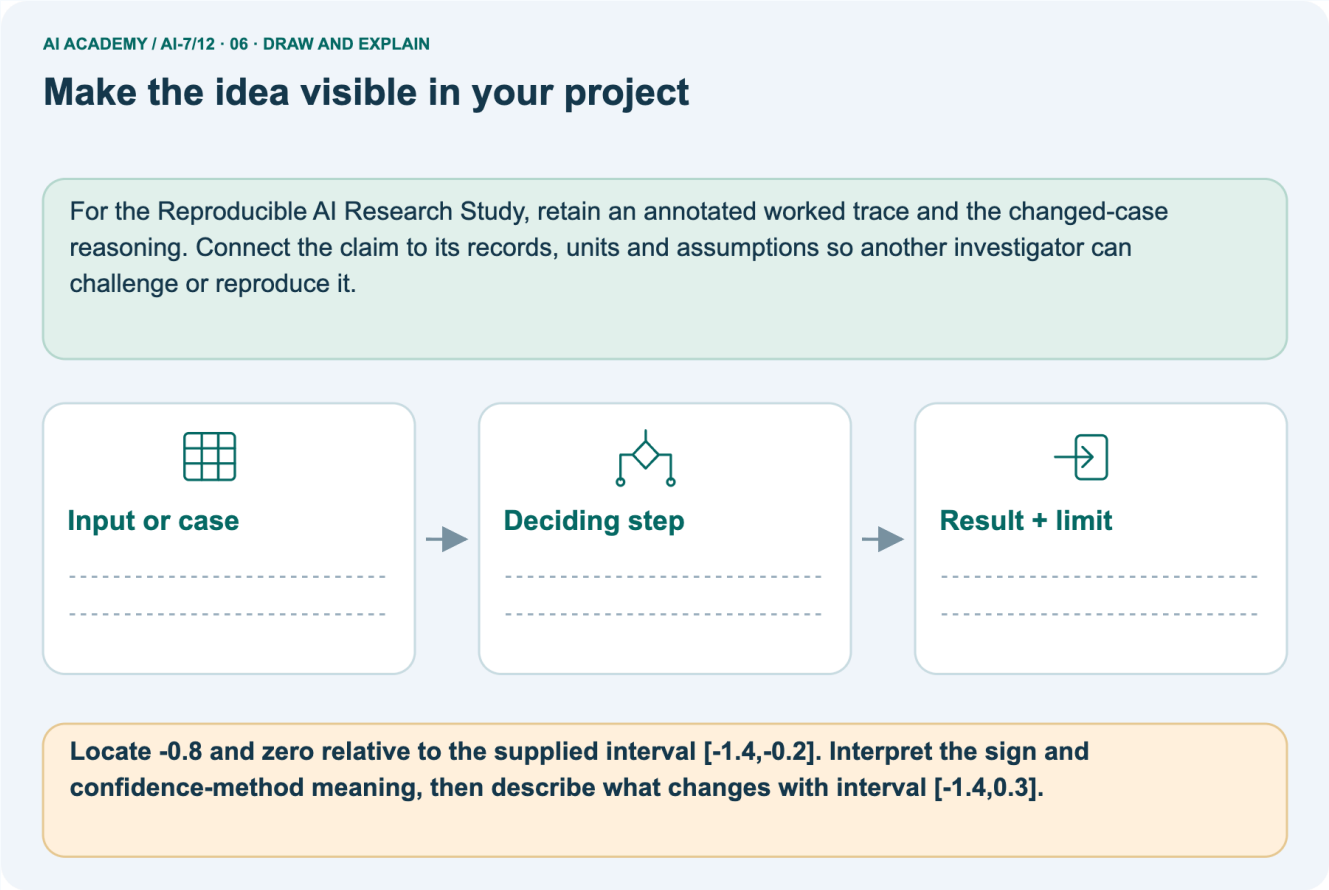
Follow the labelled picture with the matching explanation. Open the vector diagram

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Limit: An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Uncertainty intervals: the case

02. Follow the deciding steps

03. Supplied interval for B minus A time

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.